Fundamentals of Mathematics Part II Mod operator

Module I

Modular Arithmetic's

Division algorithm

- Given any positive integer *n* and any non-negative integer a, if we divide a by n, we get an integer quotient *q* and an integer remainder *r* that obey the following relationship.

$$a = q*n + r \quad , \quad 0 \le r < n \; , \; q = a/n$$

$$5 = 3*1 + 2 \text{ for } 5/3 \; .$$

$\frac{a}{n} = \frac{5}{3}$

n 3)5(1 q
3
2
r

5 = 3*1 + 2

Divisor

- A non-zero *b* divides *a* if *a* = *m***b* for some *m,* where a, b and m are Integers.
- i.e., b divides a if <u>there is no remainder</u> on division.
- The notation is commonly used to mean *b* divides *a*. Also, if b | a, we say that b is a **divisor** of a. Here, b evenly divides a.

Modular Arithmetic's

Properties of Divisor

- The following relations hold:
- If a | 1, then a $= \pm 1$.
- If a | b and b | a, then a $= \pm b$.
- Any $b \neq 0$ divides 0.
- If b | g and b | h, then $b | (mg + nh)$ for arbitrary integers *m* and *n*.

divisor Property

$\boxed{a = bm}$

$$\left.\begin{array}{l} b|g, \; g = bk \\ b|h, \; h = bk' \end{array}\right\} b \mid mg+nh$$

$$mg+nh = mbk + \cancel{m}^{n}bk'$$

$$= b(mk + nk')$$

$$mg+nh = b \cdot m$$

$$\Rightarrow b \mid mg+nh \checkmark$$